
Problem 9.19. Define the ***unique-sat*** problem to be

$$USAT = \{\langle \phi \rangle \mid \phi \text{ is a Boolean formula that has a single satisfying assignment}\}.$$

Show that $USAT \in \mathrm{P}^{SAT}$.

Proof. To show that $USAT \in \mathrm{P}^{SAT}$, we give an oracle Turing machine M^{SAT} that decides $USAT$ in polynomial time.

M^{SAT} = "On input $\langle \phi \rangle$, where ϕ is a Boolean formula:

1. Query the oracle to determine whether $\phi \in SAT$. If the oracle answers NO, then *reject*.
2. Repeat for each variable x_i in ϕ:
3. Make a new copy ϕ^T of ϕ, and replace all occurrences of x_i in ϕ^T with TRUE.
4. Query the oracle to determine whether $\phi^T \in SAT$.
5. Make a new copy ϕ^F of ϕ, and replace all occurrences of x_i in ϕ^F with FALSE.
6. Query the oracle to determine whether $\phi^F \in SAT$.
7. If the oracle answers YES both times, then *reject*.
8. *Accept*."

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